



Applications: Logarithms

Logarithmic functions have applications in many scientific fields.

Chemistry Application: pH

The pH of a substance indicates its acidity. Substances that are more acidic have a lower pH than substances that are less acidic. The logarithmic function $f(x) = -\log_{10} x$ models the pH of different substances. Specifically, the function $\text{pH} = -\log_{10} [\text{H}^+]$ measures the pH of a substance, where $[\text{H}^+]$ is the hydrogen-ion concentration of the substance, measured in moles per liter.

► **Think About It** Pure water has a pH level of 7.0, which is considered neutral on the pH scale. You can use pH paper to test whether a substance is acidic (below 7.0 on the pH scale), alkaline (above 7.0), or neutral.

EXAMPLE 1

The pH of a brand of apple juice is 3.5. What is the $[\text{H}^+]$ for the apple juice?

SOLUTION

Use the logarithmic function $\text{pH} = -\log_{10} [\text{H}^+]$. The pH level of the juice is 3.5.

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$3.5 = -\log_{10} [\text{H}^+] \quad \text{Substitute 3.5 for pH.}$$

$$-3.5 = \log_{10} [\text{H}^+] \quad \text{Multiply both sides by } -1.$$

$$[\text{H}^+] = 10^{-3.5} \quad \text{Definition of a logarithm}$$

There are $10^{-3.5}$ moles of hydrogen ions in a liter of apple juice with a pH level of 3.5. By using a calculator and rounding the result to four decimal places, you can determine that $[H^+]$ is approximately 0.0003 mole/L. ■

► **Think About It** The value in the calculator window may read 3.16227766E-4, which means $3.16227766 \times 10^{-4}$.

Biology Application: Exponential Growth

Under the right conditions, bacteria populations grow exponentially according to the formula $N_t = N_0 e^{ct}$, where N_0 is the initial number of bacteria (at time $t = 0$), t is the number of days, and N_t is the number of bacteria after t days. The constant c is the growth factor for a particular type of bacteria in a given environment.

EXAMPLE 2

At the beginning of an experiment, a culture has 400 bacteria. Three days later, the culture has 2000 bacteria.

A Calculate the growth factor.

SOLUTION

Find the value of c by substituting the given information.

$$N_0 = 400, t = 3, N_t = 2000$$

$$N_t = N_0 e^{ct}$$

$$2000 = 400 \cdot e^{c \cdot 3}$$

$$\frac{2000}{400} = e^{3c} \quad \text{Divide both sides by 400.}$$

$$5 = e^{3c}$$

$$\ln 5 = 3c \quad \text{Definition of natural logarithm}$$

$$\frac{\ln 5}{3} = c \quad \text{Divide both sides by 3.}$$

The growth factor of the function is $\frac{\ln 5}{3}$.

► **Think About It** Substitute the exact value for c . Round only at the end of the calculation.

B Predict the number of bacteria 8 days after the beginning of the experiment.

SOLUTION

Substitute the answer from Example 1A for c and 8 for t .

$$N_t = N_0 e^{ct}$$
$$N_t = 400 \cdot e^{\frac{\ln 5}{3} \cdot 8}$$

$$N_t \approx 29,240 \quad \text{Use a calculator and round to the nearest whole.}$$

There will be about 29,240 bacteria 8 days after the beginning of the experiment. ■

Physics Application: Decibels

The threshold of hearing I_0 is the intensity of the faintest sound that can be heard by the human ear. Other sound intensities I are often measured by comparing them to the threshold of hearing. The logarithmic function

$R = 10 \log_{10} \left(\frac{I}{I_0} \right)$ can be used to find the relative intensity of a sound in decibels.

EXAMPLE 3

The sound intensity of a vacuum cleaner is about 10^8 times greater than the threshold of hearing. What is the relative intensity of a vacuum cleaner, measured in decibels?

SOLUTION

Use the logarithmic function $R = 10 \log_{10} \left(\frac{I}{I_0} \right)$. Since the sound of the vacuum cleaner is 10^8 times greater than the threshold of hearing, $I = 10^8 \cdot I_0$. Substitute $10^8 \cdot I_0$ in the equation for I .

$$R = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$R = 10 \log_{10} \left(\frac{10^8 \cdot I_0}{I_0} \right) \quad \text{Substitute } 10^8 \cdot I_0 \text{ for } I.$$

$$R = 10 \log_{10} (10^8) \quad \text{Simplify by using exponential form.}$$

$$R = 10 \cdot 8$$

$$R = 80$$

The relative sound intensity of a vacuum cleaner is about 80 decibels. ■

Seismology Application: Richter Scale

The Richter scale was developed to compare the sizes of earthquakes. On the Richter scale, the magnitude of an earthquake M is related to the released energy E , in ergs, of the earthquake according to the formula

$$M = \frac{2}{3} \log_{10} \left(\frac{E}{10^{11.8}} \right).$$

EXAMPLE 4

An earthquake in the region of the Dominican Republic measured 3.2 on the Richter scale, while an earthquake near New Zealand measured 6.4. Compare the amount of energy released by these two earthquakes.

► **Think About It** An earthquake with magnitude between 3.5 and 5.4 on the Richter scale is felt, but rarely causes major damage. An earthquake with magnitude 7 or more is major and can cause serious damage.

SOLUTION

Use $M = \frac{2}{3} \log_{10} \left(\frac{E}{10^{11.8}} \right)$ to calculate the amount of energy released by each earthquake.

Dominican Republic earthquake:

$$3.2 = \frac{2}{3} \log_{10} \left(\frac{E}{10^{11.8}} \right)$$

$$4.8 = \log_{10} \left(\frac{E}{10^{11.8}} \right)$$

$$10^{4.8} = \frac{E}{10^{11.8}}$$

$$10^{4.8} \cdot 10^{11.8} = E$$

$$10^{16.6} = E$$

$$3.9810 \times 10^{16} \approx E$$

New Zealand earthquake:

$$6.4 = \frac{2}{3} \log_{10} \left(\frac{E}{10^{11.8}} \right)$$

$$9.6 = \log_{10} \left(\frac{E}{10^{11.8}} \right)$$

$$10^{9.6} = \frac{E}{10^{11.8}}$$

$$10^{9.6} \cdot 10^{11.8} = E$$

$$10^{21.4} = E$$

$$2.5119 \times 10^{21} \approx E$$

To compare the stronger quake to the weaker one, write the fraction.

$$\frac{2.5119 \times 10^{21}}{3.9810 \times 10^{16}} = \frac{2.5119}{3.9810} \times 10^5 \approx 6.31 \times 10^4$$

The 6.4 quake released roughly 63,100 times the energy of the 3.2 quake. ■

Nuclear Application: Calculating Half-Life

The function $f(t) = b \cdot \left(\frac{1}{2}\right)^{\frac{t}{h}}$ gives the amount remaining, in milligrams, of a radioactive sample of b milligrams that has a half-life h after a period of time t .

EXAMPLE 5

A sample of 50 mg of technetium-99m is given to a patient during a medical test. If the radioactive substance decays to 2.8 mg after 25 h, find the half-life of this substance.

► **Remember** The half-life of a radioactive substance is the time it takes for half of the substance to decay.

SOLUTION

$$f(t) = b \cdot \left(\frac{1}{2}\right)^{\frac{t}{h}}$$

$$2.8 = 50 \cdot \left(\frac{1}{2}\right)^{\frac{25}{h}}$$

Substitute the given values.

$$0.056 = \left(\frac{1}{2}\right)^{\frac{25}{h}}$$

Divide both sides by 50.

$$\log 0.056 = \log \left(\frac{1}{2}\right)^{\frac{25}{h}}$$

Take the common logarithm of both sides.

$$\log 0.056 = \frac{25}{h} \log \frac{1}{2}$$

Power Property of Logarithms

$$h = 25 \cdot \frac{\log \frac{1}{2}}{\log 0.056}$$

Multiply both sides by $\frac{h}{\log 0.056}$.

Use a calculator to find an approximate value of h , the half-life of this substance. Then check the result.

$$h = 25 \cdot \frac{\log \frac{1}{2}}{\log 0.056} \approx 6$$

Recall that h and t have the same units—in this case, hours.

► **Think About It** The units for h and t must be the same.

CHECK

$$2.8 = 50 \left(\frac{1}{2} \right)^{\frac{25}{h}}$$

$$2.8 \stackrel{?}{\approx} 50 \left(\frac{1}{2} \right)^{\frac{25}{6}}$$

$$2.8 \approx 2.7841 \quad \checkmark$$

The half-life of technetium-99m is approximately 6 h. ■

EXAMPLE 6

Carbon-14 has a half-life of 5730 years. Archaeologists find a wooden bowl that has 40% of the normal carbon-14 levels. When you replace b in the

half-life formula with 1, the formula $f(t) = \left(\frac{1}{2} \right)^{\frac{t}{h}}$ gives the percentage left after t years, where h is the half-life. What is the approximate age of the bowl?

SOLUTION

$$f(t) = \left(\frac{1}{2} \right)^{\frac{t}{h}}$$

$$0.4 = \left(\frac{1}{2} \right)^{\frac{t}{5730}}$$

Substitute the given values.

$$\log 0.4 = \log \left(\frac{1}{2} \right)^{\frac{t}{5730}}$$

Take the common logarithm of both sides.

$$\log 0.4 = \frac{t}{5730} \log \frac{1}{2}$$

Power Property of Logarithms

$$\frac{5730}{\log \frac{1}{2}} \cdot \log 0.4 = t$$

Multiply both sides by $\frac{5730}{\log \frac{1}{2}}$.

Use a calculator to find an approximate value of t , the age of the bowl. Then check the result.

$$t = \frac{5730}{\log \frac{1}{2}} \cdot \log 0.4 \approx 7575$$

CHECK

$$0.4 = \left(\frac{1}{2}\right)^{\frac{t}{5730}}$$

$$0.4 \approx \left(\frac{1}{2}\right)^{\frac{7575}{5730}}$$

$$0.4 \approx 0.39999 \checkmark$$

The wooden bowl is approximately 7575 years old. ■

Formula Reference

pH $\text{pH} = -\log_{10} [\text{H}^+]$

Bacteria Growth $N_t = N_0 e^{ct}$

Decibel Scale for Sound Intensity $R = 10 \log_{10} \left(\frac{I}{I_0} \right)$

Richter Scale for Earthquake Energy $M = \frac{2}{3} \log_{10} \left(\frac{E}{10^{11.8}} \right)$

Half-Life $f(t) = b \cdot \left(\frac{1}{2}\right)^{\frac{t}{h}}$